

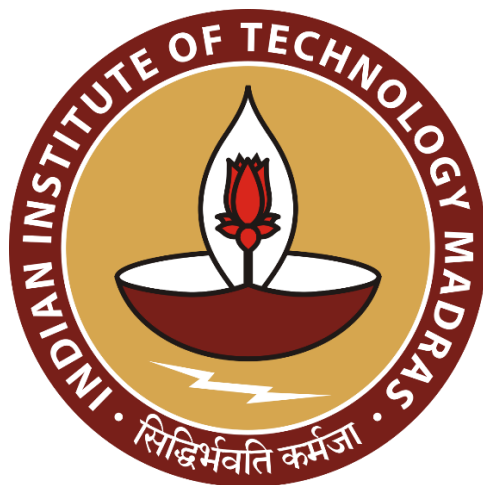
Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 1 Report

Group 9

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1. Abstract

The rise of synthetic media, such as Deep Fakes and AI-generated text, presents major risks to information integrity and digital trust. Malicious actors exploit these tools to spread disinformation, manipulate public perception, and compromise individuals' reputations. This project proposes a two-part system to address this challenge.

A vision-based classifier detects whether an uploaded image is real or manipulated using fine-tuned image model ***openai/clip-vit-base-patch32***. A text credibility pipeline evaluates user-submitted text by retrieving evidences from Google Fact Check, Tavily web-search API, and Wikimedia sources. The model ***microsoft/deberta-v3-base*** fine-tuned on the FEVER dataset classifies the claim as real or fake based on the gathered evidence, while ***mistralai/Mistral-7B-Instruct-v0.2*** produces a natural-language explanation of the final credibility assessment.

2. Objectives

- **Deep Fake Detection** – Develop an image-based classifier capable of identifying synthetic images by using deep learning models trained on labeled datasets of real and fake visuals. The system should generalize well across diverse sources, lighting conditions, and manipulation techniques.
- **Text Credibility Analysis** – Enable users to submit text and receive a detailed credibility report through Natural Language Processing and fact-checking algorithms.
- **End-to-End User Experience** – Provide a seamless and intuitive interface for users to submit both images and text, ensuring rapid and transparent feedback. The platform will emphasize usability, accessibility, and interpretability of results.

This work primarily benefits journalists combating misinformation, educators teaching media literacy, fact-checkers, and the general public who want quick verification tools.

3. Literature Review

3.1. Existing Approaches in Image Deep Fake Detection

Deepfake detection has advanced with large-scale datasets and deep learning. **FaceForensics++** introduced over 1.8 million manipulated images, enabling standardized evaluation; **XceptionNet** achieved strong detection under compression (Rössler et al., 2019). The DeepFake Detection Challenge (DFDC) released >100,000 videos, where ensemble CNNs achieved $AUC \approx 0.90$, but performance declined on unseen manipulations (Dolhansky et al., 2020).

Lightweight CNNs such as **MesoNet** (Afchar et al., 2018) and forensic methods analyzing **GAN fingerprints** (Marra et al., 2019) focus on efficiency and artifact detection. However, most models rely solely on visual cues, limiting cross-domain robustness. Recent work suggests that vision–language models like **CLIP** (Radford et al., 2021) could enhance generalization by leveraging semantic alignment between image and text.

3.2 Existing Approaches in Text Credibility and Fact-Checking

Automated fact-checking typically follows a **Retrieve–Rerank–Verify** pipeline. The **FEVER** dataset contains 185,445 claims linked to Wikipedia evidence; baseline systems achieved 31.9% accuracy, and best shared-task systems reached a **FEVER score of ~64%** (Thorne et al., 2018). Later datasets such as **FEVEROUS** (Aly et al., 2021) added tabular evidence to test multi-modal reasoning.

Modern systems use Transformer-based models like **BERT** and **DeBERTa** for claim verification (Zhou et al., 2019). Surveys highlight two open challenges—low retrieval recall and lack of explainability (Kotonya & Toni, 2020; Guo et al., 2022). **RAG** methods (Lewis et al., 2020) now combine external evidence with large language models to generate explanations, though evidence freshness and grounding remain issues.

3.3 Contribution of Proposed Solution

This work proposes an integrated multimodal credibility pipeline addressing both visual and textual misinformation:

- A **CLIP-ViT-B/32** image classifier detects manipulated images, extending beyond CNN-only detectors.
- Text claims are verified using **Google Fact Check**, **Tavily**, and **Wikimedia** retrieval, enhancing evidence diversity
- A **DeBERTa-v3-base** model fine-tuned on FEVER predicts real or fake based on retrieved evidence.
- **Mistral-7B-Instruct** generates human-readable explanations for interpretability.

Together, these components bridge gaps in prior work by combining multi-source evidence retrieval, vision–language reasoning, and explainable decision-making.

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4. Model Architecture

4.1 Deep Fake Detection (Image-Only Submission)

- **Input:** User-submitted image.
- **Model:** Pre trained Vision Model - CLIP-ViT-B/32
 - **Finetuning:** The model will be finetuned on a carefully curated dataset of "real" and "fake" images, with a focus on diverse deepfake generation methods.
 - **Output Layer:** A classification head (e.g., a fully connected layer with a sigmoid activation) will output a probability score indicating whether the image is a fake or not (e.g., 0 for real, 1 for fake).
- **Output:** Binary classification (Fake/Real).

4.2 Disinformation Detection (Text-Only Submission)

- **Input:** User-submitted text.
- **Component 1: LLM Summarization:**
 - **Function:** Extracts key claims, entities, and the core message from the input text.
 - **Prompt Engineering:** Careful prompt design will guide LLM to produce concise and accurate summaries suitable for web queries.
- **Component 2: Web Search with APIs:**
 - **Function:** Takes the summarized claims and performs targeted web searches using APIs (e.g., Tavily, Perplexica).
 - **Output:** A collection of search results (links, snippets, page content) from various sources.

- **Component 3: LLM Pro Credibility Report Generation:**
 - **Input:** Original text, summarized claims, and retrieved web search results.
 - **Function:** Gemini analyzes the provided information to:
 - Compare claims against external evidence.
 - Identify factual inconsistencies or contradictions.
 - Assess source credibility based on search results.
 - Highlight emotionally charged or biased language.
 - Provide contextual information.
 - Generate a structured report outlining the credibility of the text.
 - **Output:** A detailed credibility report, potentially including a numerical score or qualitative assessment (e.g., "Highly Credible," "Partially True," "Misleading," "False").

5. Dataset

To support the image forgery detection pipeline, we will rely on publicly available labeled image datasets that provide clear ground truth (real vs fake).

- **Kaggle:** [Deep Fake and Real Images Dataset](#)

This dataset contains a large set of labeled fake and real images, generated using GAN-based methods and paired with authentic samples. It is directly usable for binary classification (real vs fake), making it suitable for fine-tuning CLIP or other transformer-based vision models.

- **OpenForensics:** [Multi-Face Forgery Detection and Segmentation In-The-Wild](#)

Introduced by Le et al. (ICCV 2021), OpenForensics is a challenging large-scale dataset designed for multi-face forgery detection and segmentation in unconstrained environments. It includes manipulated and real faces in-the-wild, providing both image-level binary labels and pixel-level annotations. This dataset is particularly valuable because it captures realistic, social media-like conditions where multiple faces appear in a single image.

- **Web Evidence (for Text Credibility Pipeline)**

For text credibility assessment, we will not rely on a static dataset but instead use real-time search APIs (Tavily, Perplexica, Google Custom Search, Bing Web Search etc.). These APIs will retrieve external evidence, which the LLM will use to support or refute the input claims.

6. Feasibility and Compute

- **Compute Resources:** GPU-backed cloud instances (e.g., Google Colab Pro or AWS EC2). CLIP fine-tuning requires moderate resources (1–2 GPUs). LLM access will be via API (Gemini or OpenAI).
- **Implementation Feasibility:** Leveraging existing pretrained models and datasets reduces training time. The main challenge is integration and evaluation.
- **Ethical Considerations:** The tool will include disclaimers to prevent misuse, avoid overconfidence in model predictions, and ensure transparency.

7. Evaluation Metrics

- **Image Detection:** Accuracy, F1-score, ROC-AUC on benchmark datasets.
- **Text Credibility:** Alignment with fact-checking sources, human evaluation of reports, consistency across queries.
- **User Experience:** Usability feedback and response time.

8. Roles

- **Project Manager** – Coordination, documentation, milestone tracking.
- **Vision Lead** – Dataset preparation, fine-tuning CLIP, image evaluation pipeline.
- **NLP Lead** – LLM summarization, credibility report design.
- **Backend/Integration Lead** – Connecting pipelines with APIs, system architecture.
- **Frontend/UI Lead** – Interface design and user experience.

9. Timeline

- **Week 1-2:** Literature review, dataset curation, initial UI sketches.
- **Week 3-4:** Image pipeline setup, preliminary CLIP fine-tuning.
- **Week 5-6:** Text pipeline setup with Gemini API + web search integration.
- **Week 7-8:** Integration testing + Evaluation +Final report.

10. Conclusion

This milestone establishes the foundation for a unified system tackling misinformation in both visual and textual domains. By fine-tuning multimodal models for DeepFake detection and coupling LLM summarization with evidence retrieval for text credibility, the system addresses a pressing societal need. The team will continue to refine datasets, pipelines, and user interfaces in subsequent milestones.